

The Brain, with Special Reference to the Telencephalon

KARL ZILLES AND GERD REHKÄMPER

Although there are numerous reports on brain structure in *Pan* (in the following discussion *Pan* refers to *Pan troglodytes*, the common chimpanzee) and *Gorilla*, there are relatively few that deal with this subject in *Pongo*. Tiedemann (1827) appears to have been one of the first to study the brain of *Pongo*, followed by Bischoff (1870, 1876), who provided detailed information from the study of two juvenile specimens. Tilney and Riley (1928) dealt at considerable length with especially the gross anatomy of the orangutan brain. Cyto- and myeloarchitectonic details have rarely been reported (e.g., Campbell, 1905; Filimonoff, 1933; Mauss, 1908, 1911), and rigorous quantitative analyses are essentially nonexistent. There are, however, data on brain and body weights and occasionally on percentage brain composition (e.g., Bischoff, 1876; Tilney and Riley, 1928). Bauchot and Stephan (1969) used these data for their formulation of an encephalization index.

Here, we shall present new qualitative and quantitative data on the brain of *Pongo* and make comparisons with other apes as well as with *Homo* in order to delineate similarities between *Pongo* and other genera in brain structure.

MATERIALS AND METHODS

Two orang-utan brains were available for this study. The first was that of an adult female originally from Sumatra. She died in the zoological garden of Cologne and was subsequently transferred to our institution's collection. The whole animal had been fixed in 70% alcohol for several years. The brain was removed from the skull, photographed, embedded in paraffin, and cut into serial sections (20 μm). One out of each group of 20 sections was mounted and stained with cresyl fast violet for Nissl stain. The overall histological preservation is very poor, but sufficient for the delineation of basic components of

the brain (cf. pages 170–2). The volumes of these brain components were determined, using the planimetric technique of Zilles et al. (1982a) (Table 12–2).

The second brain came from the collection of the Netherlands Institute of Brain Research. It had been in different fixation fluids (the last years in formalin) over a period of 75 years. Only the telencephalon and diencephalon were present, the brainstem and cerebellum having earlier been removed. This brain was embedded by us in paraffin and cut into several sections. Instead of using a Nissl stain with cresyl fast violet, which gives poor results with old museum material, we employed a new silver staining technique, which marks the cytological details of the perikarya and the most proximal part of their process (modified after Merker, 1983). This partial brain was used for the analysis of cytoarchitecture, the delineation of different parts of the telencephalon (cf. pages 163–7), and for the determination of volumes (Table 12–2).

The structure of the different cortical areas was analyzed with a computer-controlled image analyzer (Micro-Videomat 2, C. Zeiss, F.R.G.) by measuring the gray level index (GLI). This parameter is the proportion of the profiles of stained particles (predominantly nerve cell bodies) on a measuring field of $18 \times 18 \mu\text{m}$ (for further information cf. Schleicher et al., 1978; Zilles et al., 1978a,b, 1979a,b, 1982b, 1984). The GLI was measured in neocortical areas 3 (part of the primary somatosensory cortex), 4 (the primary motor cortex), 17 (the primary visual cortex), and 41/42 (auditory cortex) by a complete scanning of a representative area (at least 6,500 measuring fields) in each cortical field. Mean values of GLIs in each cortical layer of each of the four areas were calculated. This permits interspecific comparison of the laminar structure after standardizing the actual mean GLIs of the layers to the mean GLI of the complete cortical area under consideration. The mean GLI of the complete area was defined as 1. Layer-specific GLIs

>1 were found in layers which had cell-packing densities higher than the mean of all layers of a respective area; a $GLI < 1$ indicates a lower cell-packing density. The volume proportion of each layer in a cortical area was determined by the ratio between the number of measuring fields in a layer and the total number of measuring fields in all cortical layers of the respective area.

MACROSCOPIC APPEARANCE OF THE BRAIN

Remarks on the macroscopic anatomy of the orang-utan brain can be found in Bolk (1902) and Connolly (1950). Very useful, high-quality

photographs of the orang-utan brain (as well as of other primate brains) have been published by Retzius (1906). The brain of the orang-utan is very compact; this is primarily due to the development of the telencephalon (Fig. 12-1). In dorsal view, the brain of an orang-utan is nearly egg-shaped, although the occipital poles are somewhat prominent and separated from each other by a caudally diverging interhemispherical fissure. In lateral view, the rostral part of the telencephalon appears to be especially well developed. The height of an orang-utan's brain is remarkable, reaching nearly three-quarters of its total length. The temporal lobe is well developed and its blunt tip is directed rostrally, thereby creating a protrusion of the outer contour of the brain. The cerebellum is quite prominent later-

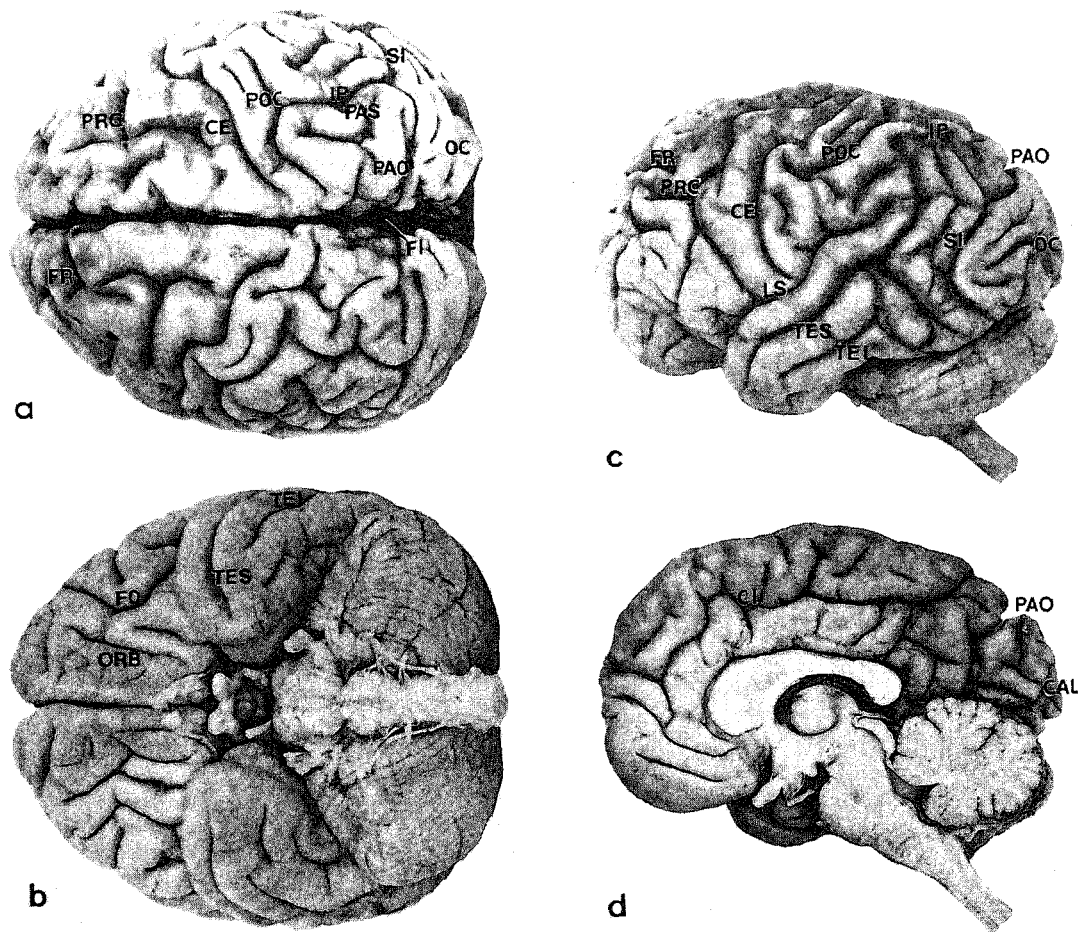


FIG. 12-1. Brain of *Pongo*. (a) Dorsal, (b) ventral, (c) lateral, and (d) medial views. CAL = calcarine sulcus; CE = central sulcus; CI = cingulate sulcus; FI = interhemispheric fissure; FO = frontoorbital sulcus; FR = frontal sulcus; IP = interparietal sulcus; LS = lateral (Sylvian) fissure; OC = occipital sulcus; ORB = orbital sulcus; PAO = parietooccipital sulcus; PAS = superior parietal sulcus; POC = postcentral sulcus; PRC = precentral sulcus; SI = simian sulcus; TEL = inferior temporal sulcus; TES = superior temporal sulcus (From Retzius, 1906).

ally and especially ventrally, although it remains inconspicuous when viewed from the dorsal aspect. A typical feature of the orang-utan brain is the keel-like form of the orbital part of the frontal lobe, which is best seen when viewed rostrocaudally (cf. Retzius, 1906). Additional special brain characteristics need to be described in comparisons with other anthropoid brains, i.e., especially between cercopithecids and the other apes.

Comparison with Cercopithecidae

Compared with cercopithecids (e.g., *Cercopithecus*, *Papio*, *Macaca*, and *Cercocebus*), the telencephalon of *Pongo* has a very elaborate sulcal pattern (Fig. 12-1). This detail is most conspicuous in the occipital pole, which is nearly smooth in cercopithecids. A significant part of the difference in the development of the orang-utan's telencephalon can be attributed to the enlargement of the frontal pole, which appears oversized when viewed laterally, in contrast to the frontal pole of a cercopithecoid, which tapers markedly and remains clearly smaller than the occipital pole. Apparently, the cortical region involved in the increased size in the rostral pole of the *Pongo* hemisphere is located much more caudally in cercopithecids. The cerebellum of *Pongo* also differs from that of cercopithecids in that it is so large, especially ventrally, so that it is nearly as broad as the entire brain. In medial aspect, the interhemispheric fissure appears to be relatively deeper in *Pongo* than in cercopithecids, the former primate also having a more elaborate sulcal pattern in its medial hemispherical surface.

Comparison with Other Apes

In features of its brain, *Pongo* appears more similar to *Gorilla* than to *Pan*. In *Pongo* and *Gorilla*, the brain in dorsal outline is egg-shaped, the thicker end lying at the occipital pole. In *Pan*, as well as in *Homo*, the anterior part of the brain is much broader in dorsal view, its rostral end being quite blunt compared to *Pongo* and *Gorilla*. In lateral view, however, some similarities between *Pongo* and *Pan* become apparent. In these two genera, the dorsal contour of the telencephalon is not semicircular, as in *Gorilla*, but exhibits a remarkable dorsal protrusion of the frontal lobe. We could not determine whether the olfactory bulbs in *Pongo* extend beyond the rostral tip of the telencephalon, as in *Gorilla*, or

if they are more truncated, as in *Pan*, because these structures were not preserved in our material and are not discernible in the illustrations published by other investigators (e.g., Retzius, 1906). Viewed ventrally, the shape of the orang-utan brain is again similar to the gorilla's. In both *Pongo* and *Gorilla*, the temporal lobe appears to be less pronounced than in *Pan*.

SULCAL PATTERN

Earlier descriptions of sulcal patterns in *Pongo* demonstrate an enormous amount of variation from individual to individual (e.g., Bischoff, 1876; Connolly, 1950; Fick, 1895a, b; Kükenthal and Ziehen, 1895; Retzius, 1906). However, some major sulci do occur regularly among different specimens and can be described using the nomenclature applicable to human brain anatomy.

The frontal lobe is caudally delineated by a central sulcus, which originates near the sulcus lateralis Sylvii and then courses toward the medial border of the hemisphere, but which does not extend onto the medial hemispherical surface (Fig. 12-1). Rostral to the central sulcus, a precentral sulcus and a frontal sulcus can be identified. These two sulci give rise to secondary sulci. In lateral and ventral views, there is an orbital sulcus and a frontoorbital sulcus (Retzius, 1906). The frontal pole is divided by several smaller sulci (e.g., the sulcus rectus of Connolly, 1950) which vary from individual to individual in shape and size.

Caudal to the central sulcus, a postcentral sulcus can be found, which can be subdivided into an inferior and a superior segment. The inferior segment may become confluent with the intraparietal sulcus, which, in turn, may be in close contact with the "sulcus simiarum" (Retzius, 1906). The homology of the "sulcus simiarum" with the sulcus lunatum in *Homo* has been the subject of much discussion (see Kuhlenbeck, 1978).

The parietal lobe exhibits an interparietal sulcus. It is further subdivided by a sulcus parietalis superior (Retzius, 1906), which runs nearly parallel to the interhemispherical fissure.

The occipital lobe displays an occipital sulcus and its derivatives, together with a very variable system of minor sulci.

Most of the temporal lobe (with the exception of the temporal pole) is separated into two parts by the long sulcus temporalis superior. The caudal and larger part of the temporal lobe bears a

sulcus temporalis inferior as well as some minor, variable sulci.

Three major sulci can be seen on the medial aspect of the brain. The first of these is the cingulate sulcus, which anteriorly may be accompanied by a parallel sulcus. The second is the parietooccipital sulcus, which crosses the medial border of the hemisphere and may contact the sulcus simiarum on the lateral aspect of the brain. The third is the calcarine sulcus, which may not even extend onto the lateral surface of the occipital lobe (Connolly, 1950; Retzius, 1906; our own observations).

CORTICAL AREAL PATTERN

Only two investigations of the cortical areal pattern of *Pongo* have been published. Campbell (1905) produced a map based on the study of cytoarchitecture, and Mauss (1908, 1911) analyzed myelin-stained material. In spite of some differences in identifying details of the borders on nonprimary cortical areas, these two studies are in nearly perfect agreement on the position and

extent of the primary areas (Figs. 12-2 and 12-3).

Mauss's (1908, 1911) area 4 and the respective area in Campbell's (1905) study can be identified as the "motor" area of Grünbaum and Sherrington (1901, 1903) and Leyton and Sherrington (1917). On the basis of physiological studies, the latter authors identified a motor area immediately rostral to the central sulcus, on the precentral gyrus, which extends over the whole length of this sulcus; motor reactions can also be elicited deep within the sulcus. Beevor and Horsley (1890-1891) had earlier reported motor reactions to electrical stimuli of *Pongo*'s cortex, but although they observed a somatotopic order, their description does not allow precise topographical identification of the stimulation sites. Leyton and Sherrington (1917) identified two additional motor regions, which lie apart from this "primary" motor field. One of these regions, on the lateral surface of the hemisphere and rostral to the precentral gyrus, leads to conjugated eye movements and thus resembles the frontal eye field in other primates, including *Homo* (Smith, 1944; Wagman, 1964). The second motor region occurs on the medial surface of the

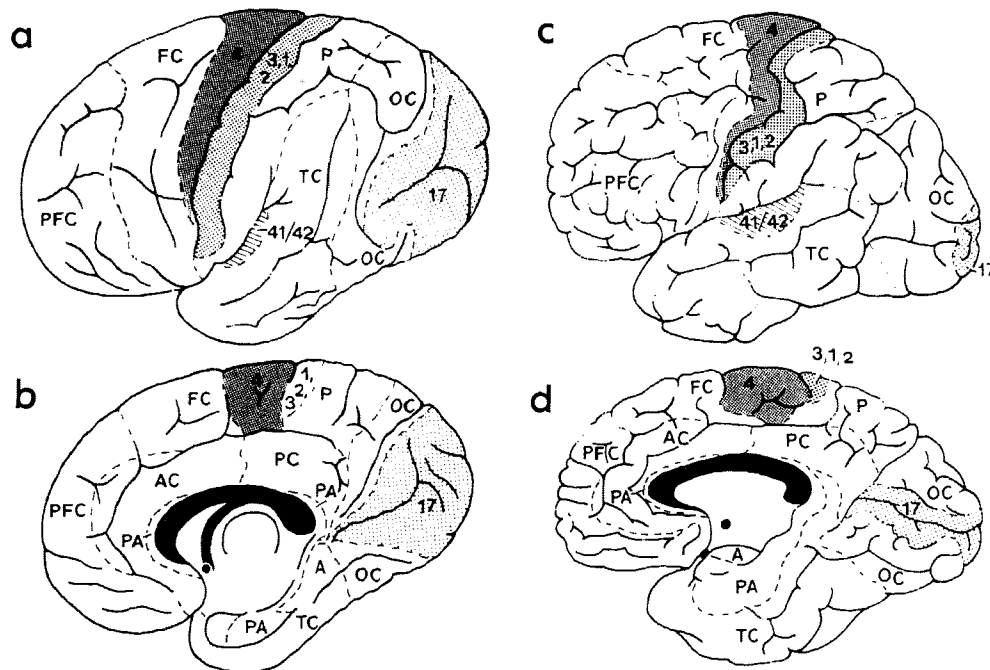


FIG. 12-2. (a) Lateral and (b) medial view of areal mapping of the cortex in *Pongo*, based on studies by Mauss (1908, 1911). (c and d) Corresponding views of cortex of *Homo*, based on studies by Brodmann (1909). 3, 1, 2 = Primary somatosensory cortex; 4 = primary motor cortex; 17 = primary visual cortex; 41/42 = auditory cortex; A = archicortex; AC = anterior cingulate cortex; FC = frontal cortex; OC = occipital cortex; P = parietal cortex; PA = periarthocortex; PC = posterior cingulate cortex; PFC = prefrontal cortex; TC = temporal cortex. All hemispheres are standardized to equal lengths of the hemispheres.

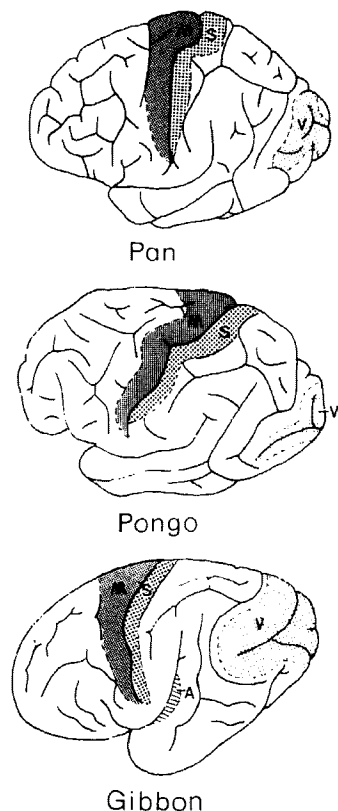


FIG. 12-3. Areal mapping of the cortex in *Pan*, *Pongo*, and gibbon (*Hylobates*) based on studies by Campbell (1905). A = auditory cortex; M = primary motor cortex; S = primary somatosensory cortex; V = primary visual cortex. All hemispheres are shown in lateral view and are standardized to equal lengths of the hemispheres.

hemisphere near the cingulate sulcus and is responsible for movements of the shoulder, trunk, wrist, and fingers. This region may be analogous to the supplementary motor area in *Homo* (Penfield and Welch, 1951).

Upon stimulation of the precentral gyrus, *Pan* and *Gorilla* show reactions similar to those of *Pongo* (Grünbaum and Sherrington, 1901, 1903; Leyton and Sherrington, 1917). Area 4 in *Pongo* is therefore comparable in position, extent, and function with area 4 in *Pan* and *Gorilla*. As such, area 4 in *Pongo* is also similar to *Homo* with respect to topography (Brodmann, 1909; Förster, 1936; von Economo and Koskinas, 1925) as well as function (Penfield and Rasmussen, 1950).

With the exception of the enormously enlarged finger representation area in *Homo*, the somatotopy in the primary motor area (Fig. 12-4) is very similar among hominoids. The topology of the different parts of the pyramidal tract is also identical, as is evident in the crura cerebri

of the hominoids (Förster 1936; Leyton and Sherrington, 1917). For example, fibers for toe and lower extremity movements are positioned most laterally in the crura cerebri. These fibers are followed medially by the fibers for the upper extremity, which, in turn, are followed by those for the fingers. The fibers for the face and tongue lie most medially.

Physiological observations of the primary sensory areas (Figs. 12-2 and 12-3) of *Pongo* have not been published. A comparison of the positions and extents of these areas in Campbell's (1905), Mauss's (1908, 1911), and Filimonoff's (1933) cortical maps of *Pongo* with those of the primary somatosensory regions (areas 3, 1, 2), visual region (area 17), and auditory regions (areas 41/42) in *Homo* (Brodmann, 1909) reveals such a striking resemblance that equivalent functions can be inferred for these areas in the orang-utan. This topological equivalence is corroborated by the cytoarchitectural comparison of their laminar pattern (see page 163). The cor-

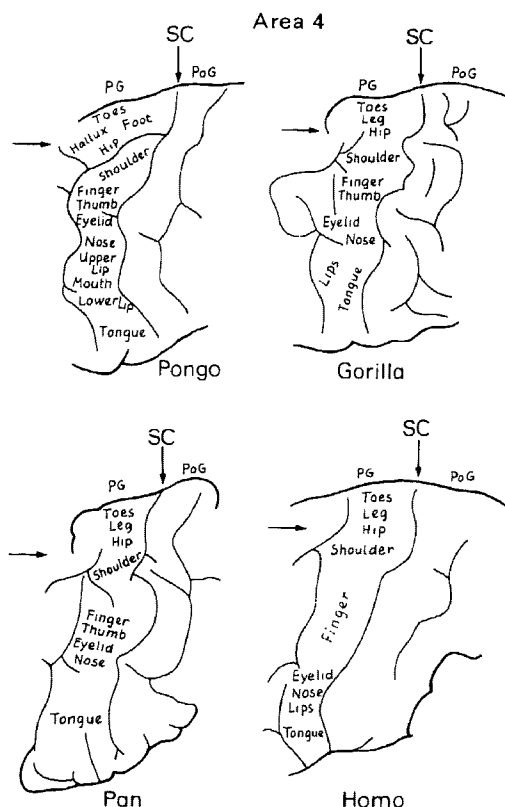


FIG. 12-4. Somatotopic representation in area 4 of *Pongo*, *Gorilla*, *Pan*, and *Homo* according to the results of Grünbaum and Sherrington (1901, 1903), Leyton and Sherrington (1917), and Penfield and Rasmussen (1950). SC = central sulcus; PG = precentral gyrus; PoG = postcentral gyrus.

tical map of *Pan* (Bailey et al., 1950; Campbell, 1905) also displays comparable topological features for the primary motor, somatosensory, and visual areas.

As a rule, the positions of the primary cortical areas in the apes and *Homo* are very similar; nevertheless, slight differences are obvious. When viewed laterally, area 17 covers only a very small part of the hemisphere in *Homo*, a greater part in *Pan* and *Gorilla*, and the greatest part in the gibbon. The situation in *Pongo* is very similar in this respect to *Pan* and *Gorilla*.

The secondary areas (Fig. 12-2) of the *Pongo* neocortex can be subdivided into many different cortical fields (Campbell, 1905; Mauss, 1908, 1911), but the methodological approaches used in these cyto- and myeloarchitectural studies, as well as the lack of any physiological analysis, do not allow for a detailed comparison of both maps in these parts of the neocortex. Therefore, we have only delineated the main groups of the secondary areas, i.e., the prefrontal cortex (PFC), the nonprimary motor cortex (FC), the parietal cortex (P), the nonprimary visual cortex (OC), and the temporal cortex (TC) (Figure 12-2).

From a structural perspective, the anterior (AC) and posterior (PC) cingulate cortices, which are found on the medial surface, although representing transitional zones between the iso- and allocortex, are more similar to the isocortex (Vogt, 1976). A very small cortical region, identified as the periarhchicortex (PA), is located between the cingulate cortex and the corpus callosum. The PA surrounds the allocortex on the medial surface of the hemisphere. The allocortex is represented in this aspect by the archicortex

(A), whose primary constituent is the hippocampus.

The paleocortex, a further part of the allocortex, is one of the smallest parts of the cerebral cortex in apes and is extremely reduced in comparison to prosimians and insectivores (Stephan, 1975). The peripaleocortex, which represents another transitional zone, is found on the lateral aspect, between the paleocortex and the isocortex.

The cyto- and myeloarchitectonical maps (Figs. 12-2 and 12-3, respectively) of the orangutan and the gibbon are also similar with regard to the position and extent of the primary cortical areas (Campbell, 1905; Mauss, 1908, 1911). Thus, this topological similarity appears as a fundamentally common structure in all primates (cf. Brodmann, 1909; Mauss, 1908, 1911; Zilles et al., 1979a,b, 1982b). Apart from this general similarity, however, some peculiarities are typical for special groups. In prosimians, the elongated primary motor and sensory areas are oriented along a roughly frontooccipital axis. The shapes of these areas are similar in platyrrhines and in lower catarrhines, but their positions are increasingly divergent from the frontooccipital axis. These areas are oriented nearly perpendicularly to the frontooccipital axis in apes and in *Homo*. Such differences may be influenced by different factors, such as skull structure, degree of encephalization (see pages 167-170), neocortical development, and formation of the central sulcus. *Pongo* is more similar to *Pan*, *Gorilla*, and *Homo* than to any other catarrhine with respect to the orientation of the primary motor and sensory areas.



FIG. 12-5. Cytoarchitecture and laminar structure of area 3 in (a) *Pongo*, (b) *Pan*, and (c) *Homo*. I = molecular layer; II = outer granular layer; III = outer pyramidal layer; IV = inner granular layer; V = inner pyramidal layer; VI = multiform layer. The magnification is identical for all examples.

LAMINAR STRUCTURE OF PRIMARY CORTICAL AREAS

The primary sensory areas (Figs. 12-5, 12-7, and 12-8) have the typical basic six-layered isocortical structure in all hominoids. This struc-

ture reaches the highest degree of differentiation in the primary visual cortex (area 17), where layer IV is divided into at least three sublayers. On the other hand, a clearly visible layer IV cannot be detected in the primary motor cortex (area 4), because this region has such a reduced

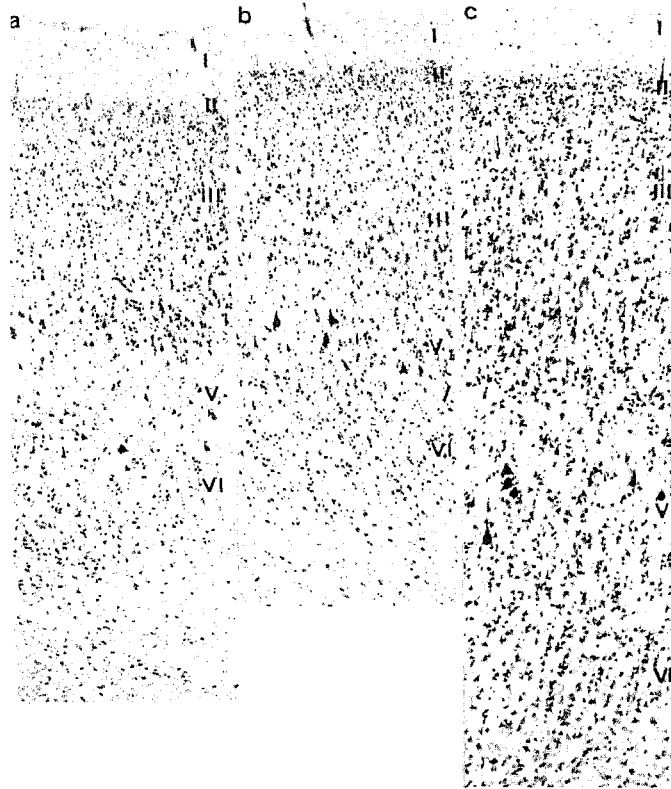


FIG. 12-6. Cytoarchitecture and laminar structure of area 4 in (a) *Pongo*, (b) *Pan*, and (c) *Homo*. For further explanation see Fig. 5.

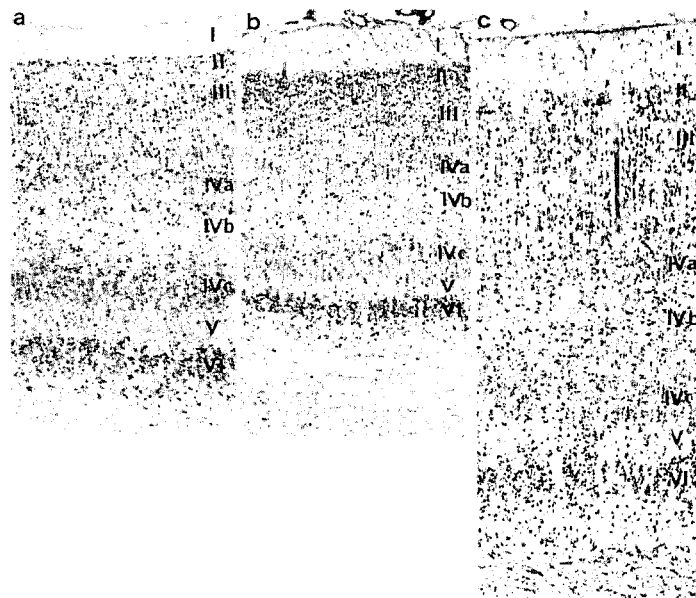


FIG. 12-7. Cytoarchitecture and laminar structure of area 17 in (a) *Pongo*, (b) *Pan*, and (c) *Homo*. For further explanation see Fig. 5.

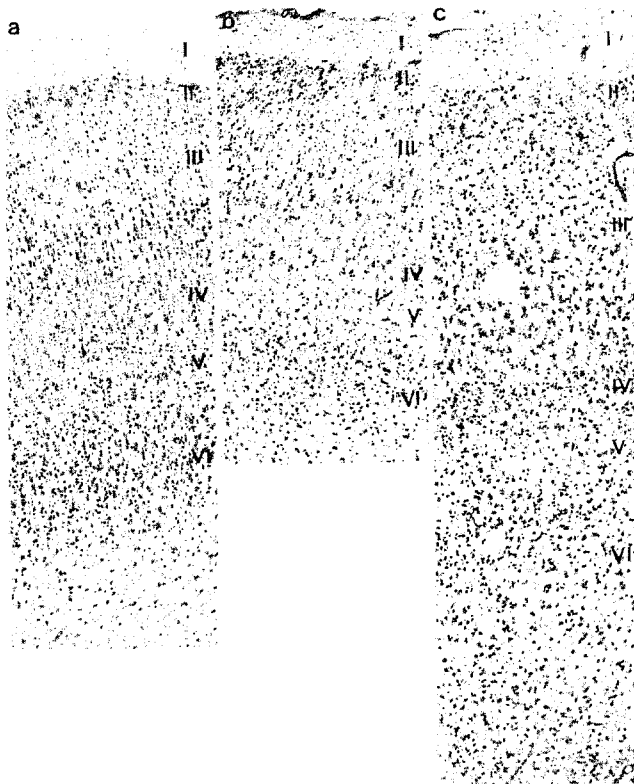


FIG. 12-8. Cytoarchitecture and laminar structure of areas 41/42 in (a) *Pongo*, (b) *Pan*, and (c) *Homo*. For further explanation see Fig. 5.

density of small granular cells (Fig. 12-6). However, since one can find the basic six-layered structure ("homotypische Formation," Brodmann, 1909), as well as a varying number of layers ("heterotypische Formation," *ibid.*), in all primates and in other mammals (Braak, 1980; Brodmann, 1909; Hubel and Wiesel, 1977; Jones, 1975; Sanides, 1968; Zilles et al., 1982b), the equivalent areas in hominoids must be distinguished with respect to several parameters, such as structural differences in cell shape and in such quantitative aspects as layer thickness and cell-packing density. Because the quantitative aspects can be objectified by measurement, we decided to use this approach for a comparison between the cortical structure in the primary areas of *Pongo*, *Gorilla*, *Pan*, and *Homo*. Since difficulties in obtaining well-preserved *Pongo* brains prohibit the analysis of a statistically sufficient sample, we attempted to minimize the effect of variation on absolute size: the variable under consideration (layer thickness or GLI as a measure of the cell-packing density) was set at 1 for the whole cortical thickness as well as for the mean GLI of each of the different cortical areas. Using this approach, the parameter can readily be interpreted as a ratio, i.e., layer thickness can be expressed as the proportion of a single layer

to the cortex in percent. The motor cortex of *Gorilla* was not available for this analysis.

The relative thicknesses of the different cortical layers (*layer proportions*) in area 3 (Fig. 12-9) show a similar pattern in *Pongo* and *Pan*. *Gorilla* and *Homo* differ in this respect, because their layer III is markedly thicker. Measurement of the same parameter in areas 4 and 17 reveals a high degree of similarity in the laminar pattern of *Pongo* and the other hominoids (Figs. 12-10 and 12-11). *Pongo* does have a higher layer IV proportion in area 41/42 than *Gorilla*, *Pan*, and *Homo*. A comparable feature is, however, evident in the overall pattern of the other layer proportions (Fig. 12-12).

The standardized GLIs of the different layers show comparable patterns in the same cortical field of apes and humans (Figs. 12-13 to 12-16). The sensory cortical areas are roughly characterized by high GLI values in layers II-IV (Figs. 12-13, 12-15, and 12-16). Area 3 in *Pongo* differs to a minor degree from the other apes and *Homo* in that, compared with layers III and IV, it has a rather low GLI in layer II and a high GLI in layers V and VI. In all large hominoids, the maximum values for the primary motor cortex (Fig. 12-14) are found in layer II followed by (in descending order) layer III, layer V, and layer

FIG. 12-9. Layer proportions in area 3 of the large hominoids (*Pongo*, *Gorilla*, *Pan*, and *Homo*) for the different cortical laminae.

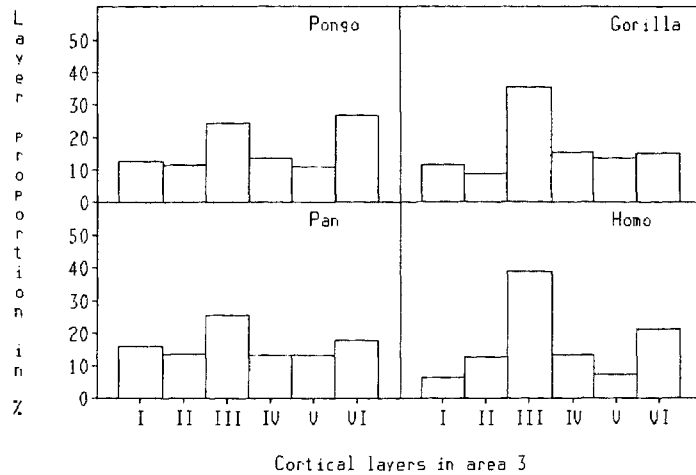


FIG. 12-10. Layer proportions in area 4 of the large hominoids (data unavailable for *Gorilla*) for the different cortical laminae.

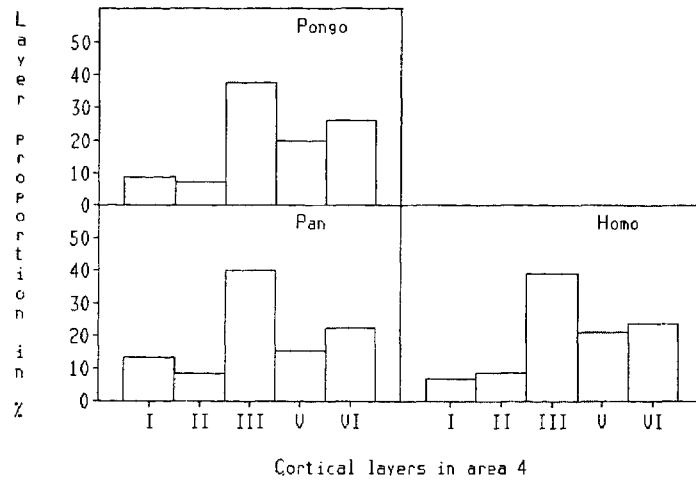
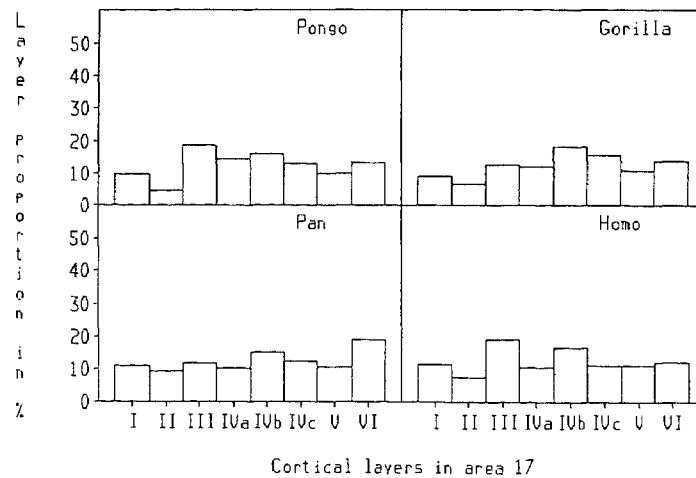


FIG. 12-11. Layer proportions in area 17 of the large hominoids for the different cortical laminae.



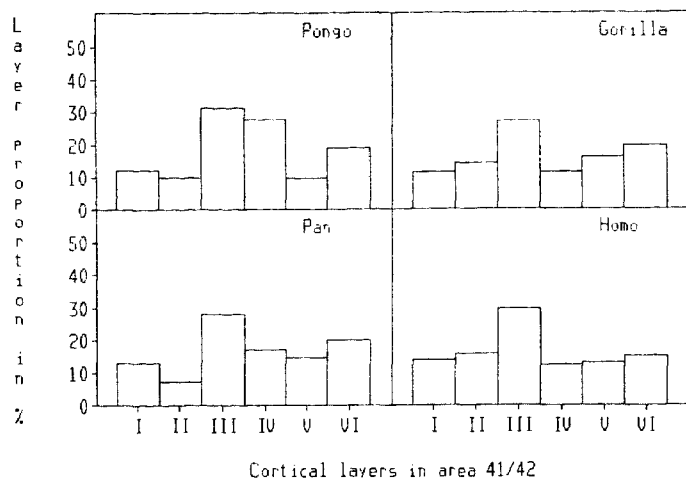


FIG. 12-12. Layer proportions in areas 41/42 of the large hominoids for the different cortical laminae.

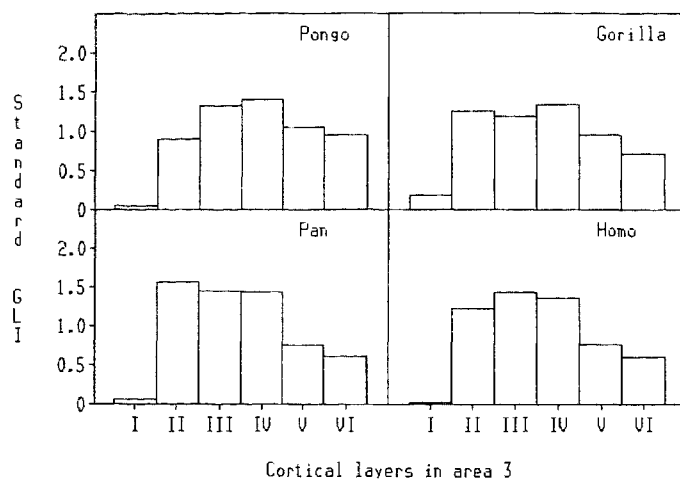


FIG. 12-13. Standardized GLIs of the different layers in area 3 of the large hominoids (*Pongo*, *Gorilla*, *Pan*, and *Homo*).

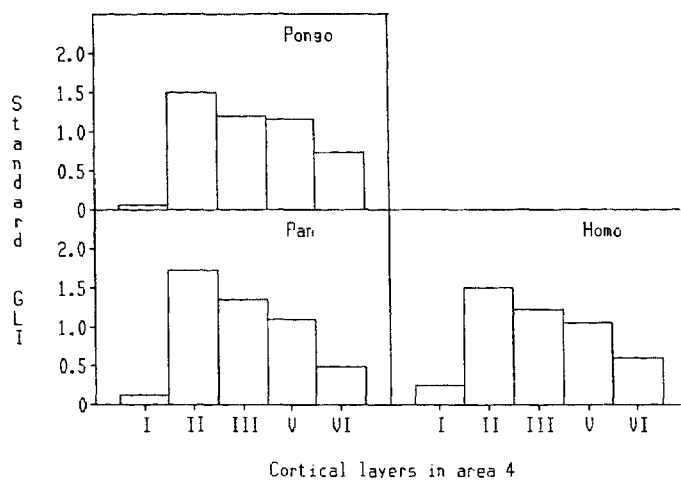


FIG. 12-14. Standardized GLIs of the different layers in area 4 of the large hominoids (data unavailable for *Gorilla*).

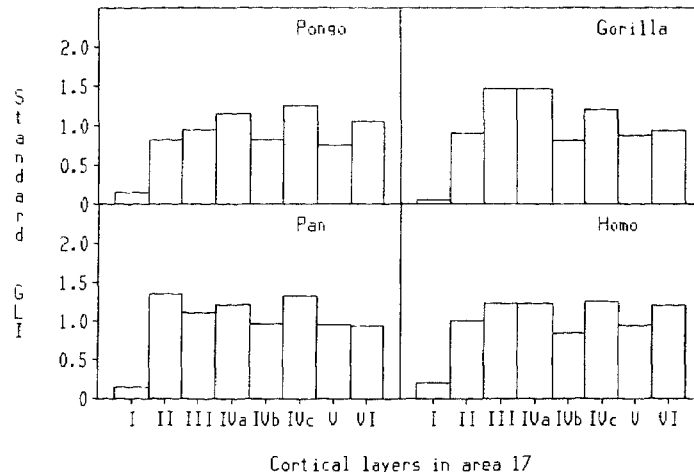


FIG. 12-15. Standardized GLIs of the different layers in area 17 of the large hominoids.

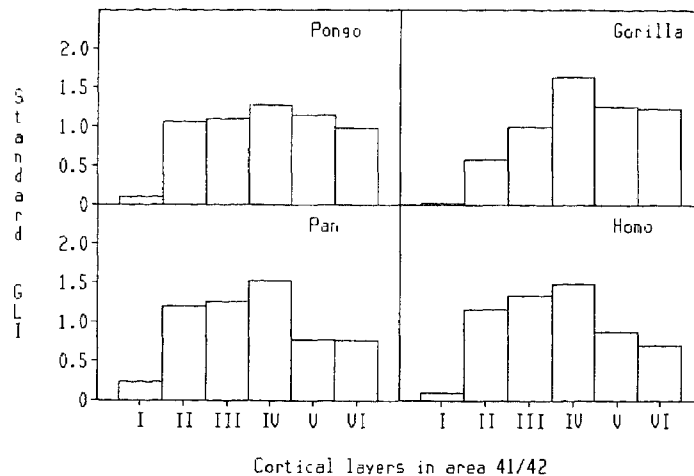


FIG. 12-16. Standardized GLIs of the different layers in areas 41/42 of the large hominoids.

VI. In summary, the cytoarchitectural structure of the primary neocortical areas is quantitatively similar in *Pongo*, *Gorilla*, *Pan*, and *Homo*, if layer thickness and cell packing density are considered as relative (standardized) values.

We can, therefore, state that the evolution of the neocortex in at least the large hominoids has led to a comparable basic structure in primary sensory cortical areas on one side and to a different laminar pattern in the primary motor areas. Our analysis of the cytoarchitectural pattern of the primary cortical areas with quantitative methods corroborates Campbell's (1905) and Brodmann's (1909) findings, which failed to uncover any fundamental differences between the apes and *Homo* in these parts of the neocortex.

ENCEPHALIZATION

The "level" of brain development, as reflected by a brain-body relationship, can be quantita-

tively expressed as an encephalization or progression index (Stephan, 1967a,b). This index is designed to reflect the relationship between the brain weight of a given species and that of a hypothetical insectivore of the same body size. Insectivores are taken as the basal comparison because insectivore-like animals are thought to most closely approximate the primitive mammalian condition and because their level of brain development is the lowest among extant mammals. The latter is particularly true of the Malagasy tenrecines, e.g., *Tenrec* and *Echinops* (Stephan, 1967c). The brain-body relationship in this tenrecid subfamily, which subsumes species of a variety of body sizes, can be described allometrically in a double logarithmic system with brain weight on the ordinate and body weight on the abscissa. In this system, the brain-body weight relation can be approximated by a regression line with a slope of 0.63 and a y -intercept of 1.6128 (Stephan et al., 1981; Stephan et al., 1986, in press). This regression line appears suitable

TABLE 12-1. Brain and Body Weights from Literature

Author	Sex	Body Weight (g)	Brain Weight (g)	Origin of the Data
Crisp 1865	j	9,157	297.7	
Deniker and Boulart 1894	?	73,000	400	Milne-Edwards?
Keith 1895	Mj	7,500	365	
	M	7,600	340.2	Manouvrier
	M	18,600	325.1	Rolleston 1861
	M	73,500	400	Owen
	F	3,170	248	Milne-Edwards
	M		283.5	
	M		425.8	Chapman 1880-81
	M		470	Owen
	F		395	Milne-Edwards
	?		288.5	Owen
Weber 1897	Mj	5,925	334.5	Owen
	M	7,600	340	
	Mj	8,830	339	
	M	11,275	375	
	M	73,500	400	
	M	76,500	395	Deniker and Boulart 1895
	F	18,593.5	315.5	Fick 1895a
	F	20,200	306	
Warncke 1908	?	18,000	333	Kohlbrugge
	Mj	7,665	365	Manouvrier
	Mj		334	Kohlbrugge
	M	53,600	400	Milne-Edwards
	M	73,200	400	Deniker and Boulart 1895
	M		283.5	Chapman 1880-81
	Mj		325.1	Owen
	M		339	Weber 1896
	Mj		340.2	Rolleston 1861
	Mj		381	
	Fj		225	Bischoff
	Fj		248	Keith
	F		306	Weber 1896
Lapicque 1907	?	73,500	400	Deniker and Boulart, also Dubois 1914, Anthony 1928, Brummelkamp 1940. See Bauchot and Stephan 1969
Filimonoff 1933	?		350	
	?		350	
	?		390	
	?		360	
Henneberg 1910	?		79.7	
Hrdlička 1925	M	49,895	347.5	
	M	54,431	371	
	M	63,503	368	
	M	72,575	326	
	M	79,379	422	
	M	83,915	384	
	M	86,183	359	
	M	90,720	395	
	F	32,659	322	
	F	32,659	326	
	F	34,019	291	

TABLE 12-1. (Continued)

Author	Sex	Body Weight (g)	Brain Weight (g)	Origin of the Data
	F	34,474	267	
	F	35,834	279	
	F	36,287	274.5	
	F	36,288	283	
	F	36,741	304	
	F	37,195	344	
	F	39,917	340.5	
	F	44,452	287.5	
Kappers 1926	?		293	See Bauchot and Stephan 1969
Kappers 1927	?		295	Weber, Deniker, and Boulart, Milne- Edwards, Fick
Putnam 1927	?		212.86	See Bauchot and Stephan 1969
	?		222.2	
Leboúcqú 1928	M		370	See Bauchot and Stephan 1969
Tilney and Riley 1928	?		246	
Bonin 1937	?	63,400	400	
Schultz 1940	M	79,300	444	
	F	1,500	129.15	
	F	9,980	359	See Bauchot and Stephan 1969
	F	36,900	343	
	F?	32,500	341	
	?	4,060	308.6	
Kennard and Willner 1941	M	4,500	325	
	M	6,700	353	See Bauchot and Stephan 1969
	F	25,200	340	
	F	30,000	315	
Count 1947	M	72,575	371.6	See Bauchot and Stephan 1969
	F	36,439	299.8	
Connolly 1950	Mj	5,415	337.5	
	Mj	7,554	284.0	Presumably the same material as Hrdlička 1925
	M	56,782	361.2	
	Fj	27,745	262.6	
	F	33,972	302.9	
Hopf 1965	M	73,000	400	
Napier and Napier 1967	M	69,000		
	F	37,000		
Walker 1975	M	75,000		
	F	40,000		

M = male; F = female; j = juvenile.

for making comparisons among orders of mammals (for methodological problems, see Stephan et al., 1986). A brain has an index of 1 if its brain-body weight relation lies on the regression line. A position above the regression line reflects an index greater than 1, while an index of less than 1 would be indicated by a position below the regression line. The progression index of the whole brain or of a region of the brain is the ratio between the mean brain weight of a species and the brain weight of a hypothetical insectivore of the same body weight as the species under consideration. The brain weight of the hypothetical insectivore can be estimated by extrapolation of the above-mentioned regression line.

Since we were unable to collect a sufficient sample of actual body weights and brain weights for *Pongo*, we had to rely on published data (see Table 12-1). These sources yield a mean body weight of 54,000 g and a fresh brain weight of 333 g. Sexual dimorphism is obvious: mean body weight in males is $72,000 \pm 14,000$ g ($N=11$) and in females is $36,000 \pm 4,000$ g ($N=14$); mean brain weight in males is 359 ± 13.9 g and in females is 306 ± 14 g. Despite these sexual differences, we worked with one mean, including data from both sexes; the means can, therefore, be regarded as representative for the species without reflecting sexual dimorphism. The progression index, which was calculated with species- but not sex-specific brain and body weights in an interspecific comparison, is thought to be reflective of the evolutionary level of brain development. Treating the sexes separately is problematic in an interspecific comparison, since this differentiation would question the evolutionary unity of sexually reproducing species. An intraspecific comparison would first have to solve the problem of defining the regression line which would be used as a reference. Many more brain-body weight pairs than have been published are required before such regression lines, which would be specific for each sex, could be calculated on a sound basis.

Potential differences in body weight (Lyon, 1911, but see Napier and Napier, 1967) between the two subspecies from Borneo and Sumatra (Jacobshagen, 1979; Röhrer-Ertl, 1982, 1983) were not considered because of the lack of precise data.

It is unclear whether the brain weight data in the literature (Table 12-1) refer to fresh weights (brains weighed immediately after death without meninges) or not. Although we could not be certain about this in any of our references, we were forced to use them for lack of better data.

Given our data base, we calculated an encephalization index (cf. page 167) of 8.48 for *Pongo* (Table 12-3), i.e., the *Pongo* brain is 8.48 times larger than the brain of a hypothetical insectivore with the same body size. Bauchot and Stephan (1969) arrived at the slightly different index of 8.90, but they relied on different brain and body weight data. We also used available data on brain and body weights (Stephan et al., 1981, in press) to calculate the encephalization index of the other extant hominoids (Table 12-3). Interestingly, the orang-utan and the gorilla emerge as having similar indices of encephalization and are, in this regard, separated from *Hylobates*, *Pan*, and *Homo*, with *Homo* being demonstrably the most distinguished. These results are in general agreement with those of Hofman (1983), who employed a different methodological approach.

DEVELOPMENT OF DIFFERENT BRAIN COMPONENTS

The different parts of the brain (see below) were delineated on photographs of serial histological sections and the areas measured with a digitizer. The volume of each brain component was calculated as outlined in Zilles et al. (1982a). We analyzed the medulla oblongata, the cerebellum (with and without the pons), the mesencephalon including tectum and ventral parts of the tegmentum (nucleus ruber, substantia nigra, nucleus subthalamicus, griseum centrale), the diencephalon, and the telencephalon. Although the delineation of components within the tegmentum is essentially artificial, such delineations are reproducible across analyses and have been followed here in order to make use of the already published data (Stephan et al., 1981). The basis for the analysis of these major brain parts has been the histological series of the brain in our institution's collection (see Materials and Methods section). The volumetric values obtained from this series must be corrected for the effect of shrinkage during histological processing. The correction factor is the quotient of the species-specific fresh volume of the brain and the brain volume in our histological series. The species-specific fresh volume of the brain is given by the species-specific brain weight of 333 g (see page 167) divided by the specific gravity of brain tissue (1.036 g/cm^3 , Stephan, 1960). This procedure not only corrects for shrinkage, but it gives brain part volumes which are more representative for

the species because it uses species-specific brain weights, rather than individual brain weights, as its basis.

Within the telencephalon, we measured the neocortex (gray plus white matter without gray of area striata), the area striata, the hippocampus, the regio entorhinalis, the paleocortex (regio prepiriformis, regio retrobulbaris, tuberculum olfactorium, and corpus amygdaloideum), the septum, and the corpus striatum (nucleus caudatus and putamen). These areas were delineated in the Netherlands brain series (see Materials and Methods section) and had to be corrected for the effect of shrinkage as well. In this case the correction factor was the ratio of the fresh volume of the telencephalon of our specimen (77.2% of total brain, see Table 12-2) and the volume of the telencephalon in the Netherlands brain series.

Table 12-2 lists the fresh volumes of the different structures in *Pongo* along with their proportion in the whole brain. The calculated indices are given in Table 12-3. The corresponding indices for *Gorilla*, *Pan*, *Hylobates*, and *Homo* were calculated from the data of Stephan et al. (1981, in press).

The index for the medulla oblongata is similar in *Pongo* and *Gorilla*. The higher values in *Pan*, *Hylobates*, and *Homo* are probably due to an enlarged pyramidal system, which passes through the medulla before entering the spinal cord.

Therefore, this result may not necessarily point out relevant differences in the function of the medulla oblongata proper, which controls various basic vegetative functions such as breathing, but it could be an indication of a more highly developed pyramidal system in *Pan*, *Hylobates*, and *Homo* as compared with *Pongo* and *Gorilla*.

The indices of the cerebellum are most similar among *Pan*, *Pongo*, and *Gorilla*, with the lowest value occurring in *Gorilla*. The index is somewhat higher in *Hylobates*, but it does not reach that of *Homo*. Since one of the main functions of the cerebellum is the coordination of motor activity, the low cerebellar volume index of *Gorilla* may well be reflective of the primarily terrestrial nature of this hominoid. Since both are actively arboreal, one would expect the index of *Pongo* to approach that of *Hylobates* most closely, but the index of *Pan* is slightly higher than the orang-utan's. The higher cerebellar index in *Pan* may, however, be related to the higher degree of neocorticalization of this hominoid (see below), since such development leads to elaborate connections of the neocortex with the cerebellum (especially the neocerebellum). Neocorticalization surely contributes in part to the high index in *Homo*.

The mesencephalic index is lowest in *Gorilla*, somewhat higher in *Pongo* and *Pan*, and highest in *Hylobates* and *Homo*. In addition to the nu-

TABLE 12-2. Volumes of Brain Components and Their Percentages of Total Brain Volume

Structure	Fresh Volume (cc ³)	Percentage of Total Brain Volume ¹
Medulla oblongata	5.5	1.8
Cerebellum (without pons)	42.9	13.9
Pons	4.3	1.4
Mesencephalon	4.0	1.3
Diencephalon	13.5	4.4
Telencephalon	238.3	77.2
Neocortex	219.8	71.2
Gray (without area striata)	129.9	42.1
Gray area striata	8.4	2.7
White matter	81.5	26.4
Hippocampus	2.7	0.9
Regio entorhinalis	1.3	0.4
Paleocortex	2.4	0.8
Regio praepiriformis	1.0	0.3
Corpus amygdaloideum	1.4	0.5
Septum	0.6	0.2
Corpus striatum	11.5	3.7
Globus pallidus	1.8	0.6

¹Excluding ventricles and nerves.

TABLE 12-3. Indices of Brain Structure Size in *Pongo* and Other Hominoidea

Structure	Slope	Y-Intercept	<i>Pongo</i>	<i>Gorilla</i>	<i>Pan</i>	<i>Hylobates</i>	<i>Homo</i>
Whole brain	0.63	1.6128	8.48	8.37	11.41	10.71	30.12
Medulla oblongata	0.60	0.7930	1.28	1.17	1.49	2.02	2.01
Cerebellum (including pons)	0.67	0.5690	8.60	8.08	8.85	9.92	22.10
Mesencephalon	0.54	0.5690	2.99	2.28	3.06	3.69	5.49
Diencephalon	0.60	0.4765	6.52	6.28	8.19	10.64	14.40
Telencephalon	0.63	1.3570	10.93	11.16	15.91	14.38	43.41
Neocortex	0.67	0.4790	49.25	49.00	72.73	66.50	199.15
Hippocampus	0.57	0.5870	1.39	1.70	2.15	5.00	4.81
Regio entorhinalis	0.60	0.0805	1.51	2.20	2.67	5.26	6.61
Paleocortex (+ corpus amygdaloideum)	0.57	1.0391	0.43	0.61	0.55	0.84	1.49
Septum	0.57	-0.0346	1.34	1.75	2.03	2.36	5.10
Corpus striatum (nucleus caudatus/putamen)	0.6	0.2570	9.18	7.83	10.80	14.77	20.56

Indices are based on a regression line through the corresponding data in Tenrecinae. Slopes and y -intercepts are from Stephan et al., 1987a; data from genera other than *Pongo* are from Stephan et al., 1981.

cleus ruber and the substantia nigra, which are parts of the extrapyramidal system, the mesencephalon contains the colliculi superior and inferior. The superior colliculus is associated with the visual system, especially the coordination of eye movement, while the inferior colliculus is associated with the auditory system; the extrapyramidal system plays a role in movements and postural adjustments. It is tempting to try to interpret differences among the hominoids in their mesencephalic indices with regard to their respective degrees of arboreality vs. terrestriality, but at present the necessary physiological data are lacking.

The orang-utan's diencephalic index is somewhat higher than in *Gorilla*, but it is clearly lower than in *Pan*; *Hylobates* and *Homo* have much higher indices. Most of the diencephalon connects the telencephalon with sensory afferents and lower centers of the brain; there are elaborate reciprocal connections with the neocortex. As indicated in other studies, the development of the diencephalon appears to be correlated with that of the telencephalon, e.g., as seen in insectivores (Stephan, 1967b,c), primates (Stephan, 1972; Stephan et al., in press), and bats (Stephan and Pirlot, 1970). This also seems to be the case in the orang-utan. A detailed qualitative description of the thalamus in *Pongo* can be found in Simma (1957) and some quantitative data in Hopf (1965) and Kraus and Gühr (1965) (see also Chapter 13, this volume).

With regard to the telencephalon, *Pongo* yields the lowest value of the hominoids, even though the difference between it and the gorilla is small; *Pan* and *Hylobates* have higher indices, and *Homo* greatly surpasses all. Brain evolution in vertebrates has generally been characterized by

an increased development of the telencephalon, a region which claims at least representative areas of all sensory systems in addition to wielding some control over all other brain functions; this enlargement is very obvious in mammals, especially primates (Stephan et al., 1981). The differences among the hominoids in telencephalic enlargement might be correlated with differences not only in diet, but in food procurement activity. For example, *Pongo* and *Gorilla* are essentially herbivorous and "lethargic" feeders, at least in comparison with the more omnivorous and "frenetic" chimpanzee; *Hylobates* is an active arborealist. The Old World monkeys may provide an analogous situation: the more "lethargic," herbivorous colobine monkeys have a lower telencephalic index than their more omnivorous cercopithecine counterparts (Stephan et al., 1981).

The allometric data on the neocortex are of interest because of the important role of this telencephalic component (Stephan, 1972). Neocortical volume has been regarded to be the most suitable parameter for defining the evolutionary level of brain development (Stephan, 1967b). In this regard, *Pongo* is just barely higher than *Gorilla* in neocortical volume, whereas *Pan* and, somewhat less so, *Hylobates* are clearly distinguished by higher indices; the index in *Homo* again surpasses all other hominoids. The primary representation areas of motor and sensory systems are located in a relatively small part of the neocortex. This part seems to decrease as the level of encephalization increases (Zilles et al., 1982b). Apparently, most of the remainder of the neocortex has the function of integrating information from different cortical areas. It can be justifiably argued that a stronger development of

the neocortex is due, perhaps primarily, to an enlargement of the integrative apparatus (Passingham, 1973; Passingham and Ettlinger, 1974; Starck, 1982), which, in turn, could be expected to be somewhat less differentiated in *Pongo* and *Gorilla* than in the other genera because of their lower indices.

The index of the hippocampus is lowest in *Pongo*, followed by *Gorilla* and then the other genera, with *Hylobates* surpassing *Homo*. Stephan (1975) and Stephan and Manolescu (1980) pointed out that the hippocampus behaves allometrically very differently among various mammals; for example, extremely high indices are found not only in *Hylobates* and *Homo*, but in *Daubentonia* and *Macroscelididae*, as well. Although the hippocampus is usually associated with learning and memory, this function may be secondary to its role in activity, attentiveness, and awareness. Since *Pongo* and *Gorilla* appear to be less active than *Pan* and *Hylobates*, it would not be unjustified to interpret *Pongo*'s and *Gorilla*'s low hippocampal indices in this context. The function of the hippocampus is complex and far from fully understood, so that interpretation of the data needs further support.

The indices of the regio entorhinalis and of the septum proceed from *Pongo* (the lowest) to *Gorilla*, *Pan*, *Hylobates*, and then *Homo* (the highest). With the exception of *Hylobates*, which has a slightly higher hippocampal index than *Homo*, the sequence of indices is the same in the regio entorhinalis and septum as in the hippocampus. The parallelism of these three areas of the limbic system results from their close interconnection (Siegel et al., 1975; Steward, 1976; for review see Stephan and Andy, 1982).

The index of the paleocortex is lowest in *Pongo*, which is followed by *Pan*, *Gorilla*, *Hylobates*, and then *Homo*. The paleocortex, as delineated here, contains part of the olfactory system as well as parts of the limbic system. Since the single components of these parts seem to have developed in very different and in partly divergent directions (Stephan, 1975; Stephan and Andy, 1977), functional interpretations are not possible until data on the single components are available.

The striatum (the nucleus caudatus and the putamen) yields a lower index in *Gorilla* than in *Pongo*; *Pongo* is followed by higher indices in *Pan* and *Homo*. The striatum appears to have undergone an enlargement from insectivores to primates (Stephan, 1979), but there are conflicting reports on the details. Harman and Carpenter (1950) suggest that the nucleus caudatus decreased in size relative to the putamen. Hopf

(1965), however, came to the opposite conclusion at least in terms of a comparison between *Pongo* and *Homo* (see also Bonin and Shariff, 1951). The striatum exhibits two main features: it is reciprocally interconnected with almost the whole neocortex and it is also an important link in the extrapyramidal system. The latter function of the striatum is of particular interest in analyzing the difference between *Pongo* and *Gorilla* in their striatum indices. The index for *Pongo* is clearly higher and lies closer to *Pan* than to *Gorilla*, and this could reflect a fundamental difference in locomotory behavior between the arboreal *Pongo* and terrestrial *Gorilla*. Differences in general among taxa in striatal indices may well be reflective of the need to control body movement in space via a well-developed extrapyramidal system.

CONCLUSIONS

In many details, the brain of *Pongo* is very similar to those of other hominoids. This is especially true with regard to cytoarchitecture and the areal arrangement of the primary areas. In particular, however, there are some features in which *Pongo* shows some special similarities primarily with *Gorilla*. For example, the brains of both genera are slender at the frontal pole in contrast to *Pan*, *Hylobates*, and *Homo*, in which the opposite condition prevails. Similarly, the indices of encephalization and the development of several brain parts, e.g., the telencephalon and the neocortex, are often as low as those of *Gorilla*, which tends to have the lowest indices of the hominoids. These latter similarities between *Pongo* and *Gorilla* probably reflect convergent solutions to similar biological demands.

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